$$\begin{aligned}5x^2+2y^2-6xy+4x-4y &= a(x-y+2)^2+b(cx+y)^2+d\\ &= (a+bc^2)x^2+(a+b)y^2+(-2a+2bc)xy+4ax-4ay+4a+d\end{aligned}$$

$$\begin{cases}a+bc^2=5\\ a+b=2\\ -2a+2bc=-6\\ 4a=4\\ 4a+d=0\end{cases} \implies \begin{cases}a=1\\ b=1\\ c=-2\\ d=-4\end{cases}$$

$$5x^2+2y^2-6xy+4x-4y=(x-y+2)^2+(y-2x)^2-4$$

$$\begin{cases}(x-y+2)^2+(y-2x)^2=13\\ (6x^2+3y^2-8xy+8x-8y)-(5x^2+2y^2-6xy+4x-4y)=5\end{cases}$$

$$\implies \begin{cases}(x-y+2)^2+(y-2x)^2=13\\ (x-y+2)^2=9\end{cases} \implies \begin{cases}x-y+2=\pm 3\\ y-2x=\pm 2\end{cases}$$

Let $u=x-y+2,\ v=y-2x$

$$\begin{cases}u=\pm 3\\ v=\pm 2\end{cases} \implies \begin{cases}x=2-u-v\\ y=4-2u-v\end{cases}$$

u	v	(x,y)
3	2	$(-3,-4)$
3	-2	$(1,0)$
-3	2	$(3,8)$
-3	-2	$(7,12)$

This solution was submitted by **Jon**

Published and refined by **Jon**

Last updated: June 25, 2026